

5(a)(i)	facilitated diffusion ; osmosis ; R osmoses / osimosis	2
5(a)(ii)	(water always moves) down a water potential gradient or from a high(er) / less negative, water potential to, low(er) / more negative, water potential ; A ψ for water potential R high water potential gradient to a low water potential gradient <i>allow ref. to the root cell has a <u>higher</u> water potential</i> <i>or adjacent cell has <u>lower</u> water potential</i>	1
5(b)(i)	endodermis ; endodermal (layer) / endodermal tissue / endodermal cells R endodermis / Casparian strip <i>giving a choice</i> A endodermis with Casparian strip <i>any one from: allow ecf from mp1</i> <i>can give descriptions of apoplast and symplast pathways instead of using terms</i> R if ref. to symplast pathway is related to movement within xylem <i>idea that water, moves from apoplast to symplast pathway (to reach xylem)</i> has to take the symplast pathway / can no longer take the apoplast pathway ; waterproof to, prevent passage of water in apoplast pathway / direct water into the symplast pathway ; protective / control, function i.e. water must cross cell surface membrane ; e.g. prevents entry of pathogens into xylem control of mineral ions	2

Question	Answer	Marks
5(b)(ii)	A xylem <u>and</u> phloem / vascular bundle, for vascular tissue <i>similarities</i> 1 endodermis surrounding vascular tissue ; 2 vascular tissue, centrally located / in the central area / not on the periphery / not near the epidermis ; I in the, middle / centre 3 epidermis with root hairs / epidermal cells with root hairs ; A outer / surface, layer with root hair cells 4 outer to inner layer described as epidermis then cortex then, endodermis / vascular tissue ; bod cortex described as parenchyma cells <i>differences</i> <i>monocotyledon</i> 5 pith / parenchyma tissue / described at the centre ; ora xylem / xylem and phloem / vascular tissue, at the centre 6 circular arrangement of vascular tissue ; A arranged in a ring 7 <i>ref. to pattern of xylem not, cross / star-shaped / AW ; context is not, e.g. tetrarch / pentarch pattern</i> 8 AVP ; cambium / cambial tissue, not visible / not present	4